Enterprise Multilingual Knowledge Assistant with Hybrid RAG & Prompt Orchestration

1. Vision

Build an enterprise AI assistant over company knowledge using RAG, Jinja2, OpenRouter, async Python and evaluation.

2. Business Problem

Employees waste time searching PDFs, policies and spreadsheets. The assistant answers from internal knowledge.

3. High-Level Architecture

User → API → NLP → Retriever → ChromaDB → Reranker → Jinja → OpenRouter → Post-processing → Evaluation.

4. Data Flow

Ingest→Clean(Pandas)→Chunk→Embed→Store→Retrieve→Prompt→LLM→Answer.

5. Sequence Diagram

User asks → Retriever searches → Context returned → Jinja renders → LLM answers.

6. Folder Structure

config/, prompts/, ingestion/, preprocessing/, rag/, vector_db/, evaluation/, cache/, logs/, tests/.

7. Why Jinja

Supports loops, conditionals, reusable templates, prompt versioning and clean separation from Python.

8. Why not PromptTemplate

Good for simple substitution but limited for production prompt orchestration.

9. Document Ingestion

Load PDF, CSV, DOCX, TXT and Excel.

10. Pandas

Cleaning, deduplication, schema normalization and analytics.

11. Chunking

Fixed, recursive, semantic and document-aware chunking.

12. Embeddings

Convert text into vectors. Compare BGE, E5 and multilingual models.

13. ChromaDB

Stores embeddings, metadata and enables similarity search.

14. Hybrid Search

Combine BM25 keyword search with vector similarity.

15. Reranking

CrossEncoder reranks Top-K retrieved chunks.

16. Query Rewriting

Improve retrieval quality before embedding.

17. NLP

Language detection, NER, keyword extraction and preprocessing.

18. Jinja Prompt Versioning

Keep prompts in prompts/*.jinja with version control.

19. Async Architecture

asyncio + aiohttp for concurrent ingestion and LLM requests.

20. OpenRouter

Use free models behind a single API interface for easy switching.

21. Caching

Cache embeddings, retrieval and responses.

22. Evaluation

Accuracy, Precision, Recall, F1, Recall@K, MRR, Faithfulness, latency and cost.

23. Logging

Track prompts, retrievals, latency and failures.

24. Production Best Practices

Environment variables, retries, rate limits, modular code and tests.

25. Week 1

Python project setup, Jinja, document loaders.

26. Week 2

Chunking and embeddings.

27. Week 3

ChromaDB and retrieval.

28. Week 4

Prompt engineering and OpenRouter.

29. Week 5

Hybrid search and reranking.

30. Week 6

Evaluation framework.

31. Week 7

Conversation memory, caching and multilingual support.

32. Week 8

Deployment, Docker, polishing and documentation.

33. Interview Preparation

Explain RAG, embeddings, vector DB, chunking, Jinja, async, evaluation metrics.

34. Final Outcome

A portfolio-quality enterprise AI assistant demonstrating modern LLM engineering.